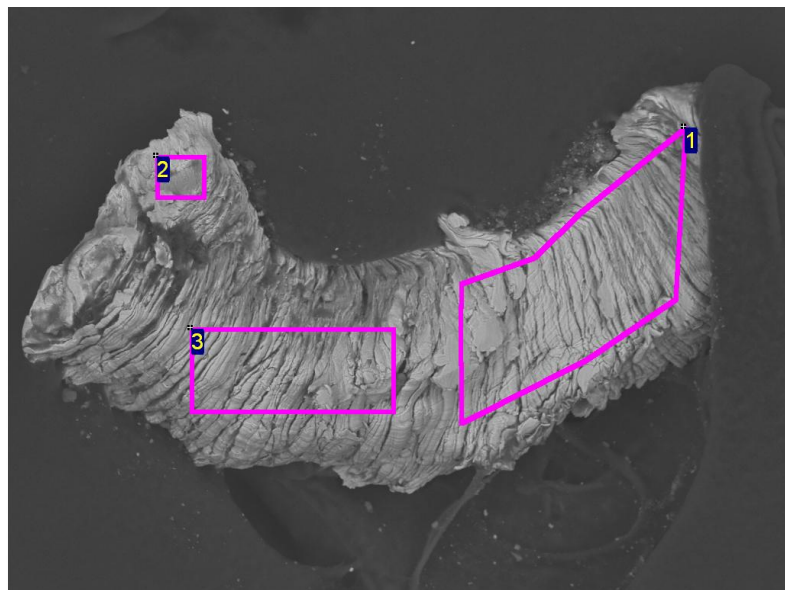


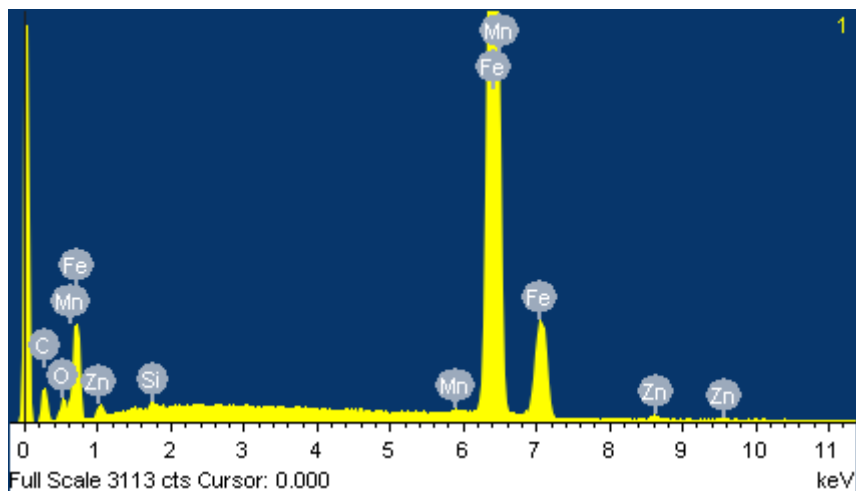
PS/QMM8-NaP		Material analysis report	
Report Date	: 04.06.2026	Report No.	: CC-CRI-QMM2-FI260552
Production		0 Km	
Investigation	✓	Field	✓
Issued By	: PS/QMM8-NaP		
Distribution:			
Subject	: Material grade and probable source identification of particles.		
1) Task	: Material analysis and dimensional measurement of 01 particles.		
2) Performed tests			
1. Scanning Electron Microscope	✓		
2. EDAX - Analysis	✓		
3. Optical Microscope	✗		
4. Hardness measurement	✗		
3) Sample preparation for SEM and EDAX:			
Particles placed on carbon tape stub.			
4) Sample preparation for microstructure investigation :			
5) Calibration			
EDAX Unit was calibrated with cobalt reference standard. EDAX analysis was performed under similar conditions with the following SEM/EDAX parameters.			
EHT: 20KV	WD: 8.5mm	Spot size :	
Signal: SE/BSD	Dead time :30-40 %	Live time : 50'	
6) Sample details			
Requisition Dept.	QMM2		
Requisition dept complaint No.	CRI.I. 26-108		
External Customer	ISUZU		
Injector no.	0445 111 115		
Mfd & Sr. no..	58021-3544		
Location of particle	In IC Stud		
Sample Submission date :		01-06-26	
Prepared by		Released by	
N R Khairnar		J R Nagrale PS/QMM8-NaP	

Project: Field CRI.I. 26-108 Particle In IC Stud (ISUZU)14
 Owner: MICO_BOSCH_NaP_QMM8
 Site: Site of Interest 1

Sample: Sample 1
 Type: Default
 ID:



Electron Image 1



Comment:

Particle Details	
Length in µm	270
Width in µm	150
Chemistry	Plain carbon steel
Surfacing coating	Traces of Zn
Probable Injector material source	Injector Body, IC stud, VSS
Probable Indirect material source at NaP	---
Probable source from pump	---
Probable external material source	---
Remarks	

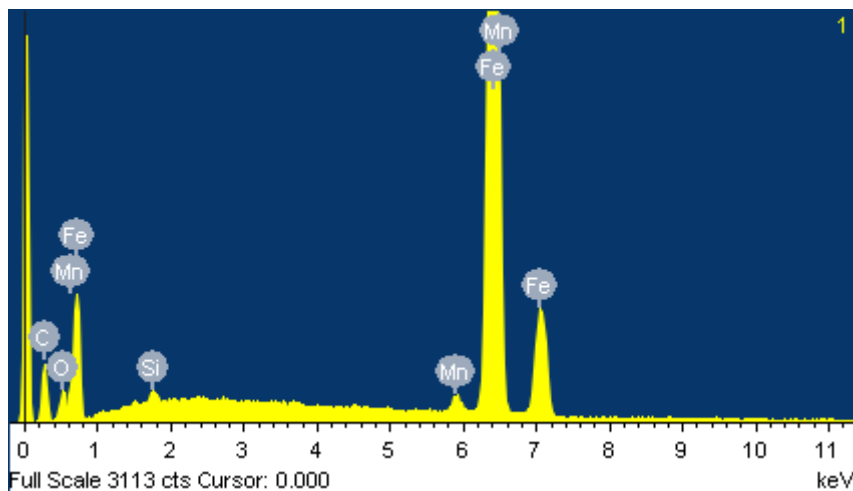
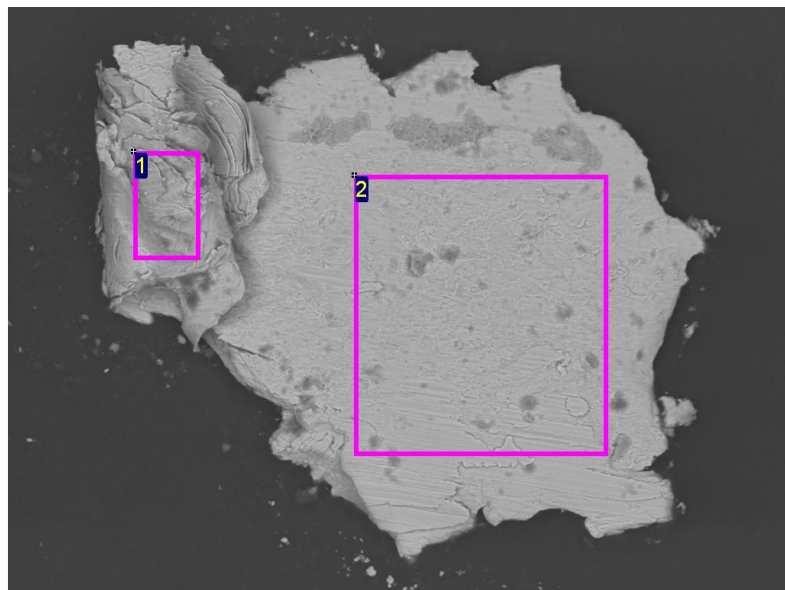
Processing option : All elements analysed (Normalised)

Spectrum	In stats.	C	O	Al	Si	Mn	Fe	Zn	Total
1	Yes	8.63	1.96		0.25	0.28	88.04	0.84	100.00
2	Yes	11.77	2.85	0.18	0.53	0.33	84.11	0.23	100.00
3	Yes	8.96	1.28			0.34	89.42		100.00
Max.		11.77	2.85	0.18	0.53	0.34	89.42	0.84	
Min.		8.63	1.28	0.18	0.25	0.28	84.11	0.23	

All results in weight%

Project: Field CRI.I. 26-108 Particle In IC Stud (ISUZU)14
 Owner: MICO_BOSCH_NaP_QMM8
 Site: Site of Interest 2

Sample: Sample 1
 Type: Default
 ID:



Processing option : All elements analysed (Normalised)

Spectrum	In stats.	C	O	Si	Mn	Fe
1	Yes	11.37	2.42	0.40	1.38	84.43
2	Yes	3.62	1.24	0.38	1.46	93.29
Mean		7.50	1.83	0.39	1.42	88.86
Std. deviation		5.48	0.83	0.01	0.05	6.27
Max.		11.37	2.42	0.40	1.46	93.29
Min.		3.62	1.24	0.38	1.38	84.43

All results in weight%

Comment:

Particle Details	
Length in μm	230
Width in μm	190
Chemistry	1.5 Mn Plain C steel
Surfacing coating	??
Probable Injector material source	??
Probable Indirect material source at NaP	BaP Line 7 Ball Guide Plate  4.2 Station 24 Shim Position  5.3 Station 40 Co-Axiality Gauge  P1 Station Cleaning area 17 Magnet Nut Spanner  Dis-Assembly Bench Cleaning area 24 Valve Set removing Base  Dis-Assembly Bench
Probable source from pump	??
Probable external material source	HPP, HPP sleeve
Remarks	